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PHY 111: Introduction to Physics

GRAVITATION (cont.)

Weight

The weight w , of a body has been defined, according to Newton's second law, as the force that causes a body of given mass m , to fall freely towards the earth with acceleration g , that is,

$$w = mg, \quad (1)$$

but in a more general term, we could also define the weight of a body as the total gravitational pull on the body by all other bodies in the universe. A body on the earth surface is mainly pulled by the earth, all other pulls say, by the moon, the sun, planet mars etc, are negligible and could be ignored.

Now, considering the earth as a uniform sphere with radius R_e and mass M_e , then the pull of the earth, that is, the weight of mass m , on the earth's surface will be given as

$$w = \frac{GM_em}{R_e^2} \quad (2)$$

Equating (1) and (2), we will have

$$mg = \frac{GM_em}{R_e^2}$$

$$g = \frac{GM_e}{R_e^2} \quad (3)$$

From Eq. 3, we could estimate the mass of earth since G , had been calculated by Cavendish in 1798, R_e has been calculated since the days of Eratosthenes and g could be calculated using the pendulum method of Galileo. Thus,

$$M_e = \frac{gR_e^2}{G}$$

using $R_e = 6380 \text{ km}$, $g = 9.80 \text{ m/s}^2$ and $G = 6.673 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$ we have

$$M_e = 5.98 \times 10^{24} \text{ kg}$$

From the fact that we assumed the earth to be a uniform spherical body and knowing its radius then we could calculate its volume as

$$V_e = \frac{4}{3}\pi R_e^3 = \frac{4}{3}\pi (6.38 \times 10^6)^3 = 1.09 \times 10^{21} \text{ m}^3.$$

Having known the mass and volume of earth, we could then go ahead and calculate the density ρ_e , of the earth, i.e.,

$$\rho_e = \frac{\text{mass}}{\text{volume}} = \frac{M_e}{V_e} = \frac{5.98 \times 10^{24}}{1.09 \times 10^{21}} \approx 5500 \text{ kg/m}^3$$

It should be noted that the density calculated here is the average density of the earth, as the density of the earth varies from the surface to the interior of the earth: at the surface the density could be about 3000 kg/m^3 while at the interior it could have a density of about 13000 kg/m^3 .

Gravitational Potential

The potential V , within the earth's gravitational field could be defined as **numerically equal to the work done in taking a unit mass from infinity to the point of observation**. By convention, the potential at infinity V_∞ , is taken as zero (0).

Now, consider a mass $m = 1 \text{ kg}$, under the influence of the earth's gravitational field with force F given by

$$F = \frac{GM}{r^2} \quad (\text{note that } m = 1 \text{ kg})$$

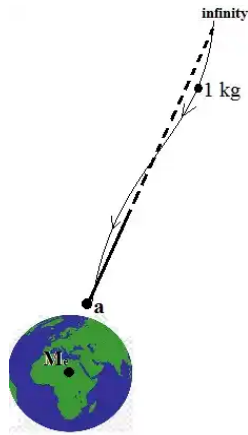


Figure 10: Object of mass 1kg moving from infinity to a point near the earth's surface.

For the work to be done on the mass m , it must move through distance δr , therefore,

$$Work = Force \times Distance$$

that is,

$$\delta V = F \times \delta r$$

$$V_a - V_\infty = \int_\infty^a \frac{GM}{r^2} dr$$

$$+V_a = -GM \left[\frac{1}{r} \right]_\infty^a = -\frac{GM}{a}$$

Since $V_\infty = 0$, the negative sign indicated that the potential at infinity is higher than the potential close to the surface. For point a , very close to the surface we can say $a \approx r$ (radius of earth), i.e.,

$$V = -\frac{GM_e}{r} \quad (4)$$

Suppose an object is on the earth's surface and we need to move this object, of mass m , far away from the earth's surface such that the object escapes from the influence of the earth's gravity, then

work done to achieve this = (mass of object) × (potential difference between infinity and the object's position)

$$\begin{aligned} &= m \times (V_\infty - V_r) \\ &= m \times \left[0 - \left(-\frac{GM_e}{r} \right) \right] \\ &= \frac{GM_e m}{r}, \end{aligned}$$

but the work done is the kinetic energy of the object as it moves away from the earth's surface, hence

$$\begin{aligned} \frac{1}{2} m v_e^2 &= \frac{GM_e m}{r} \\ v_e^2 &= \frac{2GM_e}{r}. \end{aligned}$$

Already, we know that acceleration due to gravity g , could be given as

$$g = \frac{GM_e}{r^2}$$

that is,

$$gr = \frac{GM_e}{r}$$

therefore,

$$\begin{aligned} v_e^2 &= 2gr \\ v_e &= \sqrt{2gr} \end{aligned} \quad (5)$$

v_e in Eq. 5 above is referred to as the escape velocity of an object from the earth's (or any other celestial bodies where g and r are known) surface, that is, for an object to escape the influence of gravity, then it must move at the above velocity! With $g = 9.8 \text{ ms}^{-2}$ and $r = 6.4 \times 10^6 \text{ m}$, for earth, then $v_e = 11 \text{ km/s}$.

For an object, such as a rocket, being launched from the surface of the earth, we have the scenarios shown in Figure 11. For the object having launch velocity greater than the escape velocity ($v > v_e$), the object will follow an **hyperbolic path** and completely escapes the gravitational force.

If the object is launched with a velocity equal to the escape velocity ($v = v_e$) then the object will follow a **parabolic path** and also escapes from the influence of the earth's gravity thereby,

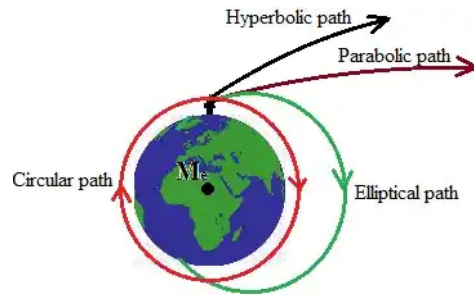


Figure 11: The path/orbit of object as it is launched from the earth's surface at different velocities.

embarking on an inter-planetary journey! For an object with launch velocity less than the escape velocity ($v < v_e$), an **elliptical path** that will eventually crash the object back to earth will be followed. At a velocity of about 8 km/s , the launched object will follow a **circular path** around the earth and if there is no friction, the velocity of the object at launch will be the same when it returns to its initial point hence, it will follow the path over and over again and remain in circular orbit around the earth: this is the path of satellites around the earth!

The concept of escape velocity could be used to explain the presence of atmosphere around some celestial bodies (e.g., earth, venus, jupiter, etc) while some others have very thin or no atmosphere at all (e.g., moon, pluto, mercury, etc).

Planetary Motion

In the solar system, the planets (Mercury, Mars, Earth, Venus, Jupiter, Saturn, Uranus and Neptune) move round the Sun using the gravitational force of attraction of the Sun. The realization that there are planets moving round the Sun as a center was made by many ancient astronomers including Tycho Brahe and Nicolaus Copernicus but it was the German astronomer, Johannes Kepler who, between 1601 and 1619, clearly described the paths of these planets as they move around the Sun.

Before Kepler's time, astronomers had used circle to describe the path of planetary motion with attendant problems, but using loads of data left by his master Tycho Brahe, Kepler was able to deduce that the path of planets around the is rather in the form of an ellipse rather than circular.

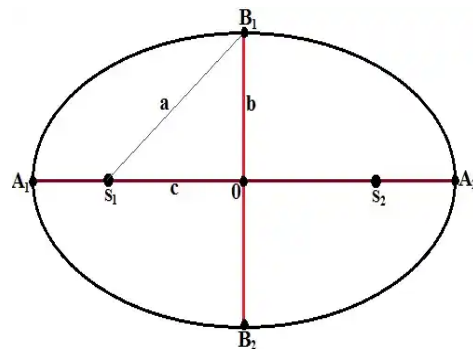


Figure 12: An Ellipse

An ellipse is a deformed circle, with one axis longer than the other, as shown in Figure 12. The longer axis is called the **major axis**, while the shorter axis is called the **minor axis**. An ellipse has two **foci**, S_1 and S_2 as shown in Figure 12. **Eccentricity e** , of an ellipse is a measure of the circularity of the ellipse, it is defined as the ratio of the distance between one focus from the center of the ellipse to the distance between the focus and the top of the semi-major axis. From Figure 12 we have,

$$\text{eccentricity, } e = \frac{OS_1}{B_1S_1} = \frac{c}{a}$$

$$0 \leq e \leq 1$$

If $e = 0$, then we have a perfect circle. The eccentricities of planets in the solar system is shown in Table 1. From the Table 1, we could see that orbits of most of the planets in the solar system are nearly circular with Venus and Neptune having almost circular orbits; Mercury has the most eccentric orbit in the solar system. The pseudo-planet, Pluto has eccentricity of about 0.248.

Table 1: Eccentricity of Planets in the Solar System

Planet	Eccentricity
Mercury	0.206
Venus	0.007
Earth	0.017
Mars	0.093
Jupiter	0.048
Saturn	0.056
Uranus	0.047

Neptune	0.009
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The motion of planets around the Sun was described by Kepler's three laws:

Law 1:

Each planet moves in an elliptical orbits with the Sun at one focus of the ellipse.

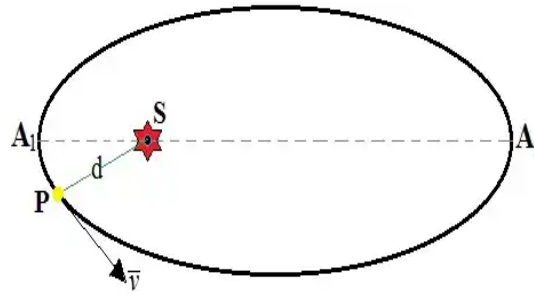


Figure 13: Kepler's First Law of Planetary Motion

At point A₁, the planet is closest to the Sun, this point is called **perihelion point**. At point A₂ the planet is farthest from the Sun, this point is called **aphelion point**.

Law 2:

A line from the Sun to a given planet sweeps out equal areas in equal times.

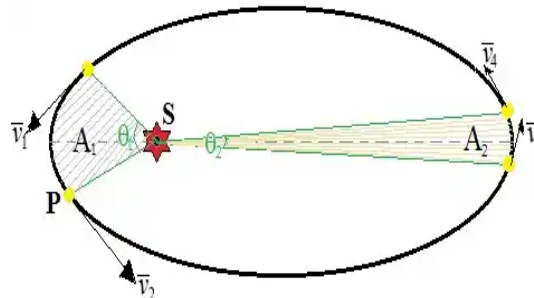


Figure 14: Kepler's Second Law of Planetary Motion

In a small time interval dt , the line from the Sun to the planet P , turns through an angle $d\theta$. The area swept out is either A_1 with r_1 or A_2 with radius r_2 . The area A , could be given as

$$dA = \frac{1}{2} r^2 d\theta$$

$$\frac{dA}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt}$$

dA/dt is called sector velocity, it is the rate at which area A , is swept out. Kepler's second law says that the sector velocity has the same value at all points in the orbit. At planet's perihelion, r is small and $d\theta/dt$ is large (the planet moves faster); at aphelion, r is large and $d\theta/dt$ is small (the planet's speed is reduced).

Law 3:

The square of the period of revolution of a planet is proportional to the cubes of its mean distance from the Sun.

The third law was announced by Kepler in 1619, apparently without knowing the real natural reason behind his finding. It was Sir Isaac Newton in 1666 who analytically was able to verify the finding of Kepler.

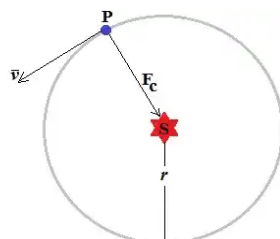


Figure 15: Newton's model of planet's orbit round the Sun.

Newton considered the motion of planet P , with mass m_p , the Sun S , with mass M_s at the center of the orbit and the radius of the orbit is r , then the centripetal force F_c , keeping the planet in its orbit is given as

$$F_c = mr\omega^2$$

but

$$\omega = \frac{2\pi}{T}$$

where ω is the angular speed of the planet and T is the period of revolution of the planet. Therefore,

$$F_c = mr \frac{4\pi^2}{T^2}$$

Between the planet and the Sun, the law of gravitational attraction holds, hence the gravitational force of attraction F_G is given as

$$F_G = k \frac{M_s m_p}{r^2}$$

hence, k is a constant. The two forces F_c and F_G must balance each other, hence,

$$F_c = F_G$$

$$m_p \frac{4\pi^2}{T^2} = k \frac{M_s m_p}{r^2}$$

$$T^2 k M_s = r^3 4\pi^2$$

$$T^2 = \left(\frac{4\pi^2}{k M_s} \right) r^3$$

From the last equation above, we could clearly see that

$$T^2 \propto r^3,$$

according to Kepler's third law.

« Gravitation	1	2	3	4	5	6	Elasticity »
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